

Quantium Virtual Internship - Retail Strategy and Analytics - Task 1

Objective

The objective of this analysis is to understand customer purchasing behaviour within the chips category and identify the customer segments that contribute most strongly to sales performance.

The analysis includes:

- data quality checks and cleaning
- feature engineering
- customer segmentation analysis
- purchasing behaviour analysis
- brand and pack size insights
- strategic recommendations for the category review

```
In [117... import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
!pip install openpyxl
```

Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)

Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)

```
In [118... try:

customer_data=pd.read_csv('/content/QVI_purchase_behaviour.csv')
transactions_data=pd.read_excel('/content/QVI_transaction_data.xlsx')
print(f'Successfully loaded the data {customer_data.shape} ')
print(f'Successfully loaded the data {transactions_data.shape} ')
except Exception as e:
    print(f'Error loading the data {e}')
```

Successfully loaded the data (72637, 3)

Successfully loaded the data (264836, 8)

Checking the Data Quality

```
In [119... transactions_data.head()
```

Out[119...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_C
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	43599	1	1307	348	66	CCs Nacho Cheese 175g	
2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	43329	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	



In [120...

```
customer_data.head()
```

Out[120...

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG SINGLES/COUPLES	Premium
1	1002	YOUNG SINGLES/COUPLES	Mainstream
2	1003	YOUNG FAMILIES	Budget
3	1004	OLDER SINGLES/COUPLES	Mainstream
4	1005	MIDAGE SINGLES/COUPLES	Mainstream

In [121...

```
transactions_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   DATE             264836 non-null int64
1   STORE_NBR        264836 non-null int64
2   LYLTY_CARD_NBR   264836 non-null int64
3   TXN_ID           264836 non-null int64
4   PROD_NBR         264836 non-null int64
5   PROD_NAME        264836 non-null object
6   PROD_QTY         264836 non-null int64
7   TOT_SALES        264836 non-null float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

In [122...

```
customer_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LYLTY_CARD_NBR        72637 non-null  int64
1   LIFESTAGE              72637 non-null  object
2   PREMIUM_CUSTOMER      72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

Missing Values

Both datasets contain no missing values.

The transaction date column is stored as an integer, so it needs to be converted into a proper date format before time-based analysis.

In [123... `transactions_data.describe()`

Out[123...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	
count	264836.000000	264836.000000	2.648360e+05	2.648360e+05	264836.000000	26
mean	43464.036260	135.08011	1.355495e+05	1.351583e+05	56.583157	
std	105.389282	76.78418	8.057998e+04	7.813303e+04	32.826638	
min	43282.000000	1.00000	1.000000e+03	1.000000e+00	1.000000	
25%	43373.000000	70.00000	7.002100e+04	6.760150e+04	28.000000	
50%	43464.000000	130.00000	1.303575e+05	1.351375e+05	56.000000	
75%	43555.000000	203.00000	2.030942e+05	2.027012e+05	85.000000	
max	43646.000000	272.00000	2.373711e+06	2.415841e+06	114.000000	

Data Cleaning and Feature Engineering

In [124... `# Convert 'DATE' to datetime objects`

```
transactions_data['DATE'] = pd.to_datetime(
    transactions_data['DATE'],
    unit='D',
    origin='1899-12-30'
)
```

In [125... `# Confirming date format conversion`

```
transactions_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                  264836 non-null  datetime64[ns]
1   STORE_NBR             264836 non-null  int64
2   LYLTY_CARD_NBR        264836 non-null  int64
3   TXN_ID                264836 non-null  int64
4   PROD_NBR              264836 non-null  int64
5   PROD_NAME             264836 non-null  object
6   PROD_QTY              264836 non-null  int64
7   TOT_SALES             264836 non-null  float64
dtypes: datetime64[ns](1), float64(1), int64(5), object(1)
memory usage: 16.2+ MB
```

In [126... transactions_data.head()

Out[126...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	

The transaction date was converted from Excel serial format into a readable date format. This allows analysis of sales trends over time, including monthly and seasonal purchasing behaviour.

Outlier Detection

Initial summary statistics identified an unusually high purchase quantity of 200 units, which require further investigation as a potential outlier.

In [127... transactions_data[transactions_data['PROD_QTY'] == 200]

Out[127...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD
69762	2018-08-19	226	226000	226201	4	Dorito Corn Chp Supreme 380g	
69763	2019-05-20	226	226000	226210	4	Dorito Corn Chp Supreme 380g	



In [128...

```
transactions_data['PROD_QTY'].value_counts()
```

Out[128...

	count
PROD_QTY	
2	236039
1	27518
5	450
3	430
4	397
200	2

dtype: int64

In [129...

```
# Remove customer who purchased 200 packets (2 rows with same LYLTY_CARD_NBR)
transactions_data = transactions_data[
    transactions_data['LYLTY_CARD_NBR'] != 226000]
```

In [130...

```
# Again checking if outlier has been removed
transactions_data['PROD_QTY'].value_counts()
```

Out[130...

	count
PROD_QTY	
2	236039
1	27518
5	450
3	430
4	397

dtype: int64

Most transactions involved between 1 and 5 chip packets, with 2-packet purchases being the most common purchasing behaviour.

Two transactions showed 200 packets purchased by the same customer, which is not

representative of normal retail shopping behaviour.

This customer was removed from the analysis to avoid distorting customer behaviour metrics.

```
In [131... transactions_data['PACK_SIZE']=transactions_data['PROD_NAME'].str.extract(r'(\d+
```

```
In [132... transactions_data[['PROD_NAME','PACK_SIZE']].head(10)
```

```
Out[132...      PROD_NAME  PACK_SIZE
```

0	Natural Chip Compny SeaSalt175g	175
1	CCs Nacho Cheese 175g	175
2	Smiths Crinkle Cut Chips Chicken 170g	170
3	Smiths Chip Thinly S/Cream&Onion 175g	175
4	Kettle Tortilla ChpsHny&Jlpno Chili 150g	150
5	Old El Paso Salsa Dip Tomato Mild 300g	300
6	Smiths Crinkle Chips Salt & Vinegar 330g	330
7	Grain Waves Sweet Chilli 210g	210
8	Doritos Corn Chip Mexican Jalapeno 150g	150
9	Grain Waves Sour Cream&Chives 210G	210

Feature Engineering

Additional product-level features were derived from the product name field to support customer purchasing analysis.

Pack size was extracted from product descriptions to analyse customer preferences across different packet sizes.

```
In [133... transactions_data['PACK_SIZE'].describe()
```

Out[133...

PACK_SIZE	
count	264834.000000
mean	182.425512
std	64.325148
min	70.000000
25%	150.000000
50%	170.000000
75%	175.000000
max	380.000000

dtype: float64

In [134...

```
transactions_data['PACK_SIZE'].value_counts().sort_index()
```

Out[134...

	count
PACK_SIZE	
70	1507
90	3008
110	22387
125	1454
134	25102
135	3257
150	43131
160	2970
165	15297
170	19983
175	66390
180	1468
190	2995
200	4473
210	6272
220	1564
250	3169
270	6285
300	15166
330	12540
380	6416

dtype: int64

Pack sizes were successfully extracted from product names to support purchasing behaviour analysis.

The most frequently purchased pack sizes were 175g and 150g, indicating strong customer preference for medium-sized chip packets.

Pack sizes ranged from 70g to 380g, with an average pack size of approximately 182g.

In [135...

```
transactions_data['BRAND'] = transactions_data['PROD_NAME'].str.split().str[0]
```

In [136...

```
transactions_data['BRAND'].head(15)
```


Out[136...

BRAND

0	Natural
1	CCs
2	Smiths
3	Smiths
4	Kettle
5	Old
6	Smiths
7	Grain
8	Doritos
9	Grain
10	Smiths
11	Kettle
12	Twisties
13	WW
14	Thins

dtype: object

In [137...

```
transactions_data['BRAND'].value_counts().sort_values(ascending=True)
```

Out[137...

	count
BRAND	
French	1418
NCC	1419
Sunbites	1432
GrnWves	1468
Burger	1564
Snbts	1576
Cheetos	2927
Smith	2963
Infzns	3144
Dorito	3183
Woolworths	4437
CCs	4551
Cheezels	4603
Red	5885
Natural	6050
Grain	6272
Tyrrells	6442
Old	9324
Twisties	9454
Tostitos	9471
Cobs	9693
WW	10320
Infuzions	11057
RRD	11894
Thins	14075
Doritos	24962
Pringles	25102
Smiths	28860
Kettle	41288

dtype: int64

```
In [138... transactions_data['BRAND'] = transactions_data['BRAND'].replace({
    'Dorito': 'Doritos',
    'Smith': 'Smiths',
    'Red': 'RRD',
    'WW': 'Woolworths',
    'Infzns': 'Infuzions',
    'Snbts': 'Sunbites',
    'GrnWves': 'Grain Waves',
    'Grain': 'Grain Waves',
    'NCC': 'Natural'
})

In [139... transactions_data['BRAND'].value_counts().sort_values(ascending=True)
```

Out[139...

	count
BRAND	
French	1418
Burger	1564
Cheetos	2927
Sunbites	3008
CCs	4551
Cheezels	4603
Tyrrells	6442
Natural	7469
Grain Waves	7740
Old	9324
Twisties	9454
Tostitos	9471
Cobs	9693
Thins	14075
Infuzions	14201
Woolworths	14757
RRD	17779
Pringles	25102
Doritos	28145
Smiths	31823
Kettle	41288

dtype: int64

Brand names were also extracted and standardised to support brand-level customer analysis.

```
In [140...] merged_data=pd.merge(transactions_data,
                        customer_data,
                        on='LYLTY_CARD_NBR',
                        how='left')
```

```
In [141...] merged_data.head(10)
```

Out[141...	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
5	2019-05-19	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	
6	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
7	2019-05-16	4	4196	3539	24	Grain Waves Sweet Chilli 210g	
8	2018-08-20	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	
9	2018-08-18	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	



```
In [142...] merged_data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264834 entries, 0 to 264833
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                  264834 non-null  datetime64[ns]
1   STORE_NBR             264834 non-null  int64
2   LYLTY_CARD_NBR        264834 non-null  int64
3   TXN_ID                264834 non-null  int64
4   PROD_NBR              264834 non-null  int64
5   PROD_NAME             264834 non-null  object
6   PROD_QTY              264834 non-null  int64
7   TOT_SALES             264834 non-null  float64
8   PACK_SIZE             264834 non-null  int64
9   BRAND                 264834 non-null  object
10  LIFESTAGE              264834 non-null  object
11  PREMIUM_CUSTOMER      264834 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(6), object(4)
memory usage: 24.2+ MB

```

```
In [143... merged_data[merged_data['LIFESTAGE'].isna()].shape
```

```
Out[143... (0, 12)
```

Merging Transaction and Customer Data

Transaction data was merged with customer segmentation data using the loyalty card number as the common key.

This combined dataset allows analysis of purchasing behaviour across different customer lifestage and premium customer segments.

```

In [144... segment_sales = merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
)['TOT_SALES'].sum().sort_values(ascending=False)

segment_sales

```

Out[144...

TOT_SALES

LIFESTAGE	PREMIUM_CUSTOMER	
OLDER FAMILIES	Budget	168363.25
YOUNG SINGLES/COUPLES	Mainstream	157621.60
RETIREES	Mainstream	155677.05
YOUNG FAMILIES	Budget	139345.85
OLDER SINGLES/COUPLES	Budget	136769.80
	Mainstream	133393.80
	Premium	132263.15
RETIREES	Budget	113147.80
OLDER FAMILIES	Mainstream	103445.55
RETIREES	Premium	97646.05
YOUNG FAMILIES	Mainstream	92788.75
MIDAGE SINGLES/COUPLES	Mainstream	90803.85
YOUNG FAMILIES	Premium	84025.50
OLDER FAMILIES	Premium	80658.40
YOUNG SINGLES/COUPLES	Budget	61141.60
MIDAGE SINGLES/COUPLES	Premium	58432.65
YOUNG SINGLES/COUPLES	Premium	41642.10
MIDAGE SINGLES/COUPLES	Budget	35514.80
NEW FAMILIES	Budget	21928.45
	Mainstream	17013.90
	Premium	11491.10

dtype: float64

In [145...

```

segment_customers = merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
)['LYLTY_CARD_NBR'].unique().sort_values(ascending=False)

segment_customers

```

Out[145...

LYLTY_CARD_NBR

LIFESTAGE	PREMIUM_CUSTOMER	
YOUNG SINGLES/COUPLES	Mainstream	8088
RETIREES	Mainstream	6479
OLDER SINGLES/COUPLES	Mainstream	4930
	Budget	4929
	Premium	4750
OLDER FAMILIES	Budget	4675
RETIREES	Budget	4454
YOUNG FAMILIES	Budget	4017
RETIREES	Premium	3872
YOUNG SINGLES/COUPLES	Budget	3779
MIDAGE SINGLES/COUPLES	Mainstream	3340
OLDER FAMILIES	Mainstream	2831
YOUNG FAMILIES	Mainstream	2728
YOUNG SINGLES/COUPLES	Premium	2574
YOUNG FAMILIES	Premium	2433
MIDAGE SINGLES/COUPLES	Premium	2431
OLDER FAMILIES	Premium	2273
MIDAGE SINGLES/COUPLES	Budget	1504
NEW FAMILIES	Budget	1112
	Mainstream	849
	Premium	588

dtype: int64

In [146...

```
segment_metrics = merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
).agg({
    'TOT_SALES': 'sum',
    'LYLTY_CARD_NBR': 'nunique',
    'TXN_ID': 'count'
})
```

In [147...

```
segment_metrics['TXN_PER_CUSTOMER'] = (
    segment_metrics['TXN_ID'] /
    segment_metrics['LYLTY_CARD_NBR']
)
```

In [148...

```
segment_metrics.sort_values(
    by='TXN_PER_CUSTOMER',
    ascending=False
)
```

Out[148...

		TOT_SALES	LYLTY_CARD_NBR	TXN_ID	TXN_I
LIFESTAGE	PREMIUM_CUSTOMER				
OLDER FAMILIES	Mainstream	103445.55	2831	14244	
	Budget	168363.25	4675	23160	
	Premium	80658.40	2273	11190	
YOUNG FAMILIES	Budget	139345.85	4017	19122	
	Premium	84025.50	2433	11563	
	Mainstream	92788.75	2728	12907	
OLDER SINGLES/COUPLES	Premium	132263.15	4750	17754	
	Budget	136769.80	4929	18407	
	Mainstream	133393.80	4930	18318	
MIDAGE SINGLES/COUPLES	Mainstream	90803.85	3340	11874	
RETIREEES	Budget	113147.80	4454	15201	
	Premium	97646.05	3872	13096	
MIDAGE SINGLES/COUPLES	Premium	58432.65	2431	8216	
	Budget	35514.80	1504	5020	
RETIREEES	Mainstream	155677.05	6479	21466	
NEW FAMILIES	Mainstream	17013.90	849	2325	
	Premium	11491.10	588	1589	
	Budget	21928.45	1112	3005	
YOUNG SINGLES/COUPLES	Mainstream	157621.60	8088	20854	
	Budget	61141.60	3779	9242	
	Premium	41642.10	2574	6281	



Customer frequency analysis shows that Older Families across all premium tiers average approximately 5 transactions per customer during the period.

In contrast, Young Singles/Couples have the largest customer base but lower purchase frequency, indicating that their contribution comes primarily from customer volume rather than repeat purchasing behaviour.


```
In [149... segment_metrics['TOTAL_UNITS'] = merged_data.groupby(  
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']  
)[ 'PROD_QTY' ].sum()
```

```
In [150... segment_metrics['UNITS_PER_TXN'] = (  
    segment_metrics['TOTAL_UNITS'] /  
    segment_metrics['TXN_ID']  
)
```

```
In [151... segment_metrics.sort_values(  
    by='UNITS_PER_TXN',  
    ascending=False  
)
```

```
Out[151...          TOT_SALES  LYLTY_CARD_NBR  TXN_ID  TXN_I
```

LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES	LYLTY_CARD_NBR	TXN_ID	TXN_I
OLDER FAMILIES	Mainstream	103445.55	2831	14244	
	Budget	168363.25	4675	23160	
	Premium	80658.40	2273	11190	
YOUNG FAMILIES	Budget	139345.85	4017	19122	
	Mainstream	92788.75	2728	12907	
	Premium	84025.50	2433	11563	
OLDER SINGLES/COUPLES	Premium	132263.15	4750	17754	
	Budget	136769.80	4929	18407	
MIDAGE SINGLES/COUPLES	Mainstream	90803.85	3340	11874	
OLDER SINGLES/COUPLES	Mainstream	133393.80	4930	18318	
RETIREEES	Premium	97646.05	3872	13096	
	Budget	113147.80	4454	15201	
MIDAGE SINGLES/COUPLES	Budget	35514.80	1504	5020	
	Premium	58432.65	2431	8216	
RETIREEES	Mainstream	155677.05	6479	21466	
NEW FAMILIES	Premium	11491.10	588	1589	
	Mainstream	17013.90	849	2325	
	Budget	21928.45	1112	3005	
YOUNG SINGLES/COUPLES	Mainstream	157621.60	8088	20854	
	Premium	41642.10	2574	6281	
	Budget	61141.60	3779	9242	

Transaction analysis indicates that units per transaction is broadly consistent across all segments at approximately 1.9, suggesting pack quantity is not a major differentiator..

Older Families, particularly Budget customers, are strong contributors to chip sales due to both high purchase frequency and larger basket sizes.

```
In [152... segment_metrics['AVG_PRICE_PER_UNIT'] = (  
    segment_metrics['TOT_SALES'] /  
    segment_metrics['TOTAL_UNITS']  
)
```

```
In [153... segment_metrics.sort_values(  
    by='AVG_PRICE_PER_UNIT',  
    ascending=False  
)
```

Out[153...

		TOT_SALES	LYLTY_CARD_NBR	TXN_ID	TXN_I
LIFESTAGE	PREMIUM_CUSTOMER				
YOUNG SINGLES/COUPLES	Mainstream	157621.60	8088	20854	
MIDAGE SINGLES/COUPLES	Mainstream	90803.85	3340	11874	
NEW FAMILIES	Mainstream	17013.90	849	2325	
	Budget	21928.45	1112	3005	
RETIREEES	Budget	113147.80	4454	15201	
	Premium	97646.05	3872	13096	
OLDER SINGLES/COUPLES	Premium	132263.15	4750	17754	
NEW FAMILIES	Premium	11491.10	588	1589	
OLDER SINGLES/COUPLES	Budget	136769.80	4929	18407	
RETIREEES	Mainstream	155677.05	6479	21466	
OLDER SINGLES/COUPLES	Mainstream	133393.80	4930	18318	
MIDAGE SINGLES/COUPLES	Premium	58432.65	2431	8216	
YOUNG FAMILIES	Budget	139345.85	4017	19122	
	Premium	84025.50	2433	11563	
MIDAGE SINGLES/COUPLES	Budget	35514.80	1504	5020	
OLDER FAMILIES	Budget	168363.25	4675	23160	
	Mainstream	103445.55	2831	14244	
YOUNG FAMILIES	Mainstream	92788.75	2728	12907	
OLDER FAMILIES	Premium	80658.40	2273	11190	
YOUNG SINGLES/COUPLES	Premium	41642.10	2574	6281	
	Budget	61141.60	3779	9242	



Average price analysis shows that Mainstream Young Singles/Couples pay the highest price per unit, suggesting a stronger preference for premium or higher-value chip products.

In contrast, family segments purchase chips more frequently and in larger quantities, but at lower average prices per unit, indicating more value-oriented purchasing behaviour.

```
In [154... brand_sales = merged_data.groupby(  
    ['LIFESTAGE', 'BRAND']  
)[ 'TOT_SALES' ].sum().unstack()
```

```
In [155... brand_sales.head()
```

```
Out[155... 
```

	BRAND	Burger	CCs	Cheetos	Cheezels	Cobs	Doritos	French	Gra Wav
	LIFESTAGE								
	MIDAGE SINGLES/COUPLES	660.1	1717.8	1499.7	4080.6	6995.8	22725.60	654.0	4919
	NEW FAMILIES	161.0	352.8	325.3	998.1	2059.6	6693.55	168.0	1423
	OLDER FAMILIES	1566.3	3830.4	3686.2	7002.9	12027.0	41822.90	1632.0	9583
	OLDER SINGLES/COUPLES	1271.9	3385.2	3390.6	8310.0	14812.4	50985.45	1566.0	10752
	RETIREEES	1131.6	2933.7	2785.5	7561.8	13687.6	46772.40	1269.0	9763

5 rows × 21 columns



```
In [156... merged_data.groupby('BRAND')[ 'TOT_SALES' ].sum().sort_values(ascending=False)
```

Out[156...

TOT_SALES	
BRAND	
Kettle	390239.8
Doritos	240590.9
Smiths	224660.2
Pringles	177655.5
Infuzions	99047.6
RRD	95046.0
Old	90785.1
Thins	88852.5
Twisties	81522.1
Tostitos	79789.6
Cobs	70569.8
Tyrrells	51647.4
Grain Waves	51617.2
Woolworths	49343.6
Natural	42318.0
Cheezels	40029.9
CCs	18078.9
Cheetos	16884.5
Sunbites	9676.4
French	7929.0
Burger	6831.0

dtype: float64

In [157...

```
top_segments = merged_data[
    (
        (merged_data['LIFESTAGE'] == 'OLDER FAMILIES') &
        (merged_data['PREMIUM_CUSTOMER'] == 'Budget')
    )
    |
    (
        (merged_data['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES') &
        (merged_data['PREMIUM_CUSTOMER'] == 'Mainstream')
    )
]
```

Visualisation of Customer and Brand Insights

The following charts summarise key customer purchasing behaviours and highlight the most commercially important customer segments and brands within the chips category.

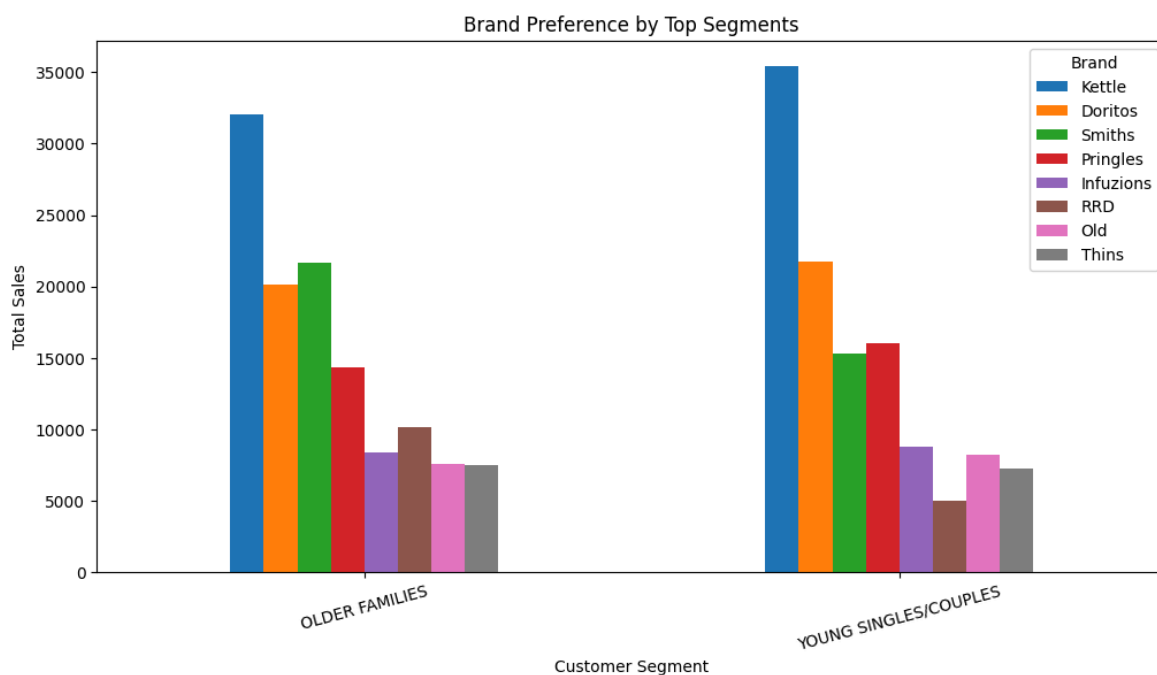
```
In [158... segment_brand_plot = top_segments.groupby(
    ['LIFESTAGE', 'BRAND']
)['TOT_SALES'].sum().unstack().fillna(0)

# Keep only top 8 brands by total sales to keep chart readable
top_brands = merged_data.groupby('BRAND')['TOT_SALES'].sum().nlargest(8).index
segment_brand_plot = segment_brand_plot[top_brands]

segment_brand_plot.plot(kind='bar', figsize=(12, 6))

plt.title('Brand Preference by Top Segments')
plt.xlabel('Customer Segment')
plt.ylabel('Total Sales')
plt.xticks(rotation=15)
plt.legend(title='Brand', bbox_to_anchor=(1, 1))

plt.savefig('brand_preference_by_segment.png', bbox_inches='tight')
plt.show()
```



Kettle dominates sales in both segments but Young Singles/Couples over-index on it, consistent with their higher spend per unit. Older Families spread spend more evenly across Smiths and Doritos compared with some other brands. Push Kettle to younger shoppers, multi-buy Smiths/Doritos deals to families.

```
In [159... merged_data['MULTIPACK'] = merged_data['PROD_QTY'] > 1
```

```
In [160... multipack_segment = merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
)['MULTIPACK'].mean().sort_values(ascending=False)
```

```
In [161... # Convert MultiIndex or tuple index into readable labels
multipack_segment_plot = multipack_segment.copy()
multipack_segment_plot.index = [
```

```

f"{life.title()} | {premium.title()}"
for life, premium in multipack_segment_plot.index
]

# Sort values
multipack_segment_plot = multipack_segment_plot.sort_values()

plt.figure(figsize=(12, 8))

ax = multipack_segment_plot.plot(kind="barh")

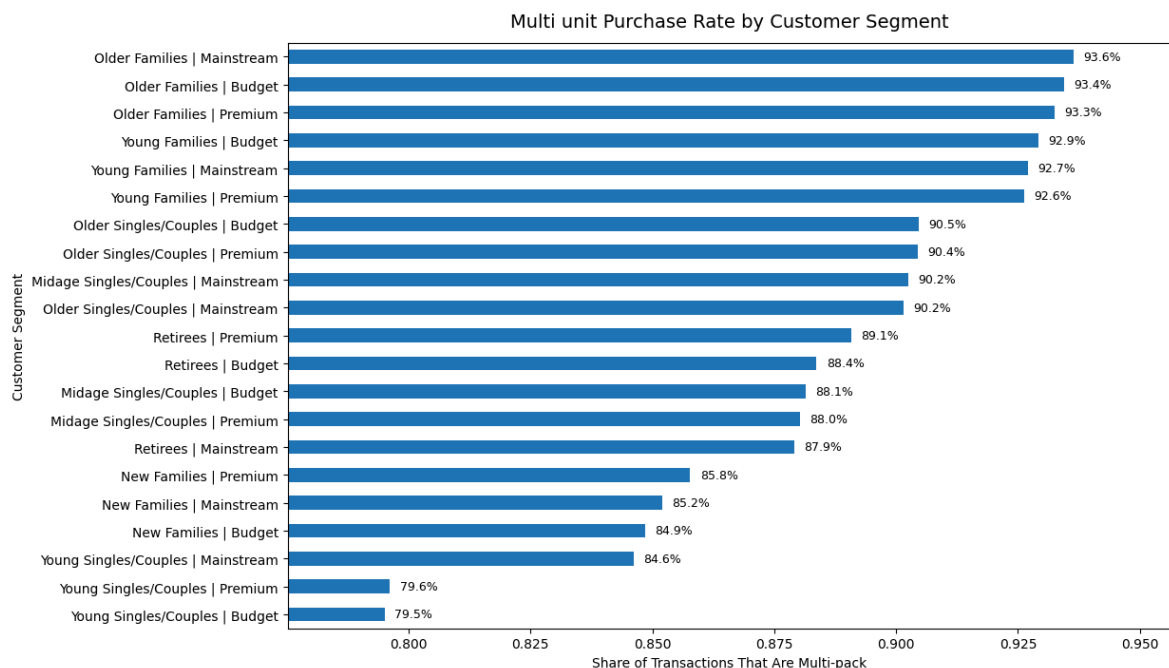
plt.title("Multi unit Purchase Rate by Customer Segment", fontsize=14, pad=12)
plt.xlabel("Share of Transactions That Are Multi-pack")
plt.ylabel("Customer Segment")

# Zoom into useful range
plt.xlim(
    multipack_segment_plot.min() - 0.02,
    multipack_segment_plot.max() + 0.02
)

# Add value labels
for i, v in enumerate(multipack_segment_plot):
    ax.text(v + 0.002, i, f"{v:.1%}", va="center", fontsize=9)

plt.savefig("multipack_rate.png", dpi=300, bbox_inches="tight")
plt.show()

```



Most customer segments frequently purchased multiple chip packs per transaction. Older Families showed the highest multi-pack purchasing behaviour, supporting their strong contribution to overall sales.

In [162...

```

segment_sales_plot = merged_data.groupby(
    ['LIFESTAGE', 'PREMIUM_CUSTOMER']
)['TOT_SALES'].sum().sort_values(ascending=False)

colors = ['#1f4e79'] + ['#b0c4de'] * (len(segment_sales_plot) - 1)
plt.figure(figsize=(12,6))

```

```

ax=segment_sales_plot.plot(kind='barh',color=colors)

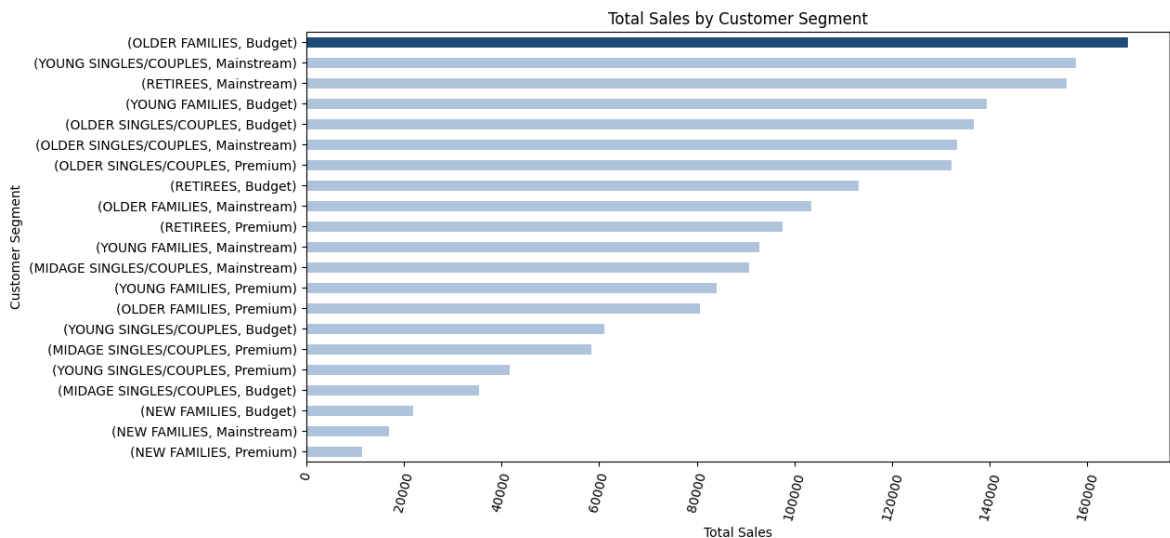
#put highest at top
ax.invert_yaxis()

plt.title('Total Sales by Customer Segment')
plt.ylabel('Customer Segment')
plt.xlabel('Total Sales')

plt.xticks(rotation=75)

plt.savefig('sales_by_segment.png', bbox_inches='tight')
plt.show()

```



```

In [163... txn_plot = segment_metrics['TXN_PER_CUSTOMER'].sort_values(ascending=False)

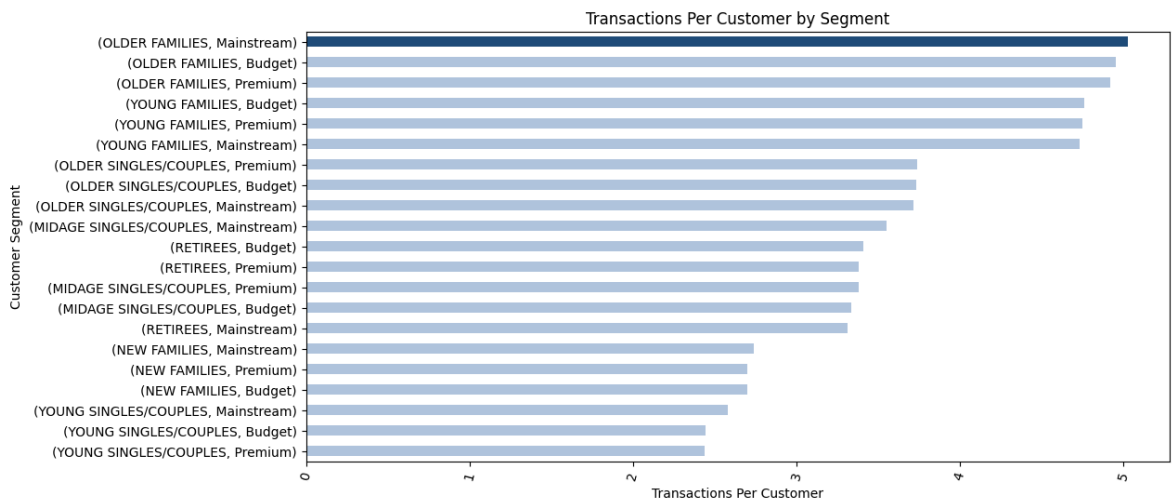
colors = ['#1f4e79'] + ['#b0c4de'] * (len(segment_sales_plot) - 1)
plt.figure(figsize=(12,6))

tx=txn_plot.plot(kind='barh',color=colors)
tx.invert_yaxis()

plt.title('Transactions Per Customer by Segment')
plt.xlabel('Transactions Per Customer')
plt.ylabel('Customer Segment')

plt.xticks(rotation=75)
plt.tight_layout
plt.savefig('transactions_per_customer.png', bbox_inches='tight')
plt.show()

```

In [164...

```
brand_plot = merged_data.groupby(
    'BRAND'
)['TOT_SALES'].sum().sort_values(ascending=False)

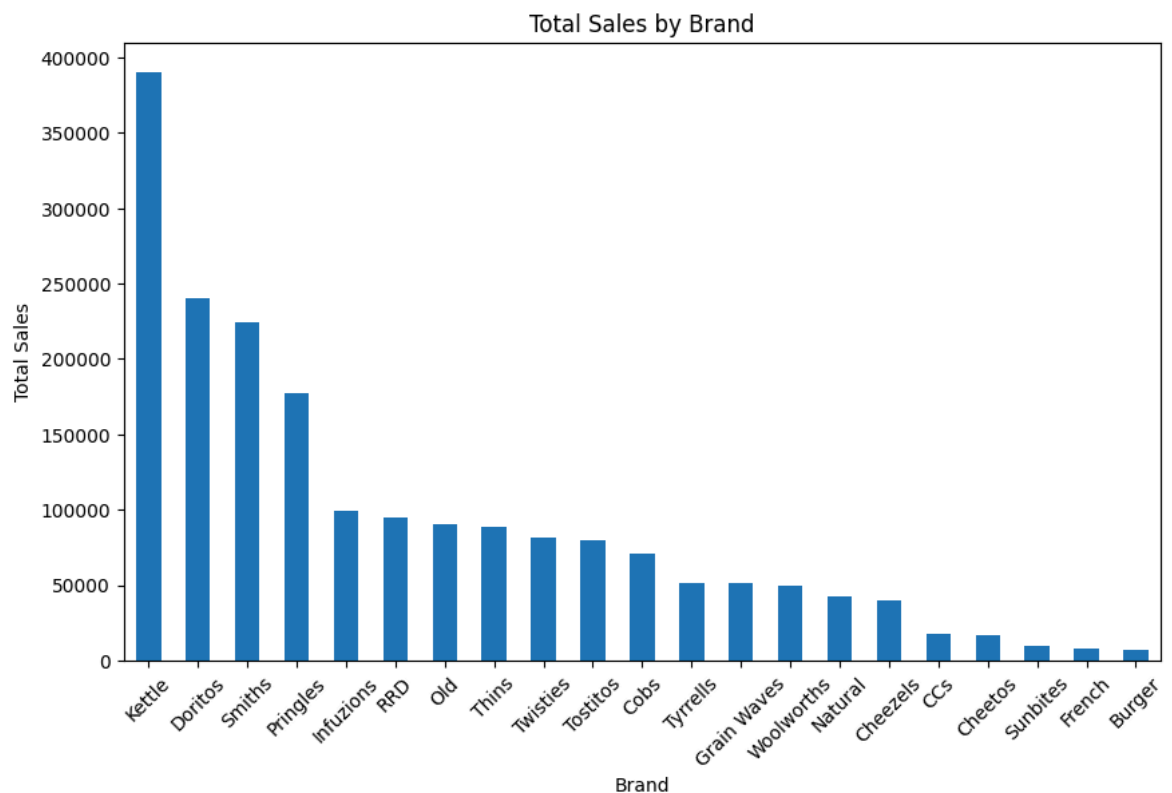
plt.figure(figsize=(10,6))

brand_plot.plot(kind='bar')

plt.title('Total Sales by Brand')
plt.ylabel('Total Sales')
plt.xlabel('Brand')

plt.xticks(rotation=45)

plt.savefig('brand_sales.png', bbox_inches='tight')
plt.show()
```



In [165...

```
monthly_sales = merged_data.groupby(
    merged_data['DATE'].dt.to_period('M')
```

```

)['TOT_SALES'].sum()

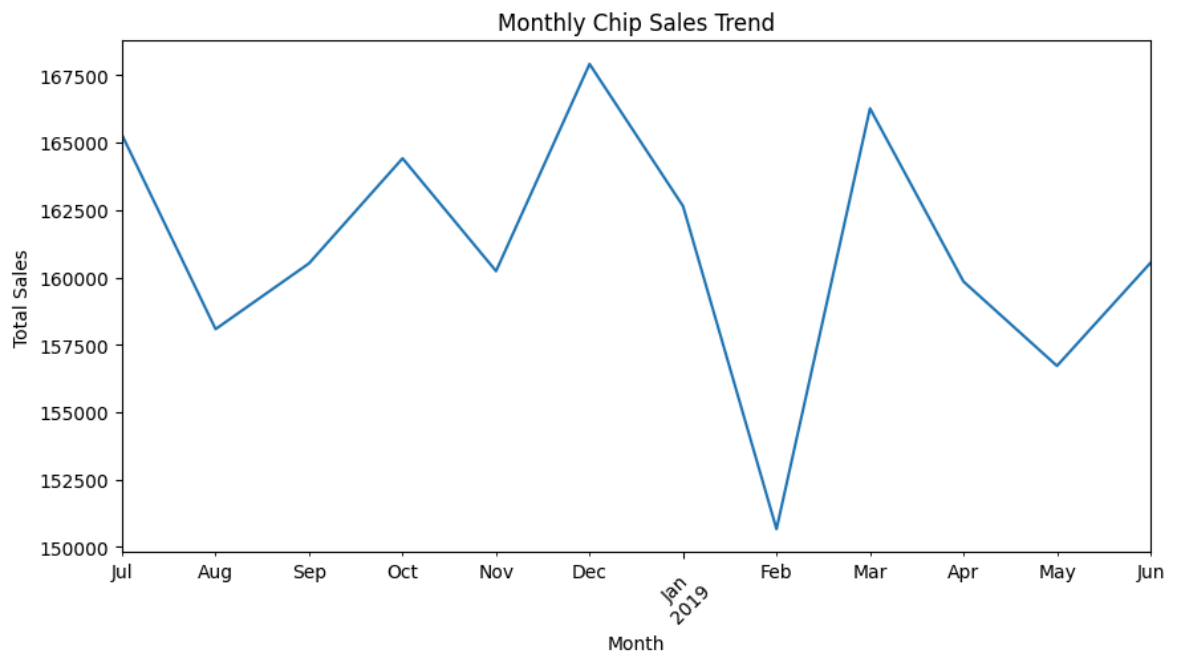
monthly_sales.plot(figsize=(10,5))

plt.title('Monthly Chip Sales Trend')
plt.ylabel('Total Sales')
plt.xlabel('Month')

plt.xticks(rotation=45)

plt.savefig('monthly_sales_trend.png', bbox_inches='tight')
plt.show()

```



Chip sales remained relatively stable throughout the year, with a noticeable peak in December and a decline in February.

The December increase may reflect seasonal holiday purchasing behaviour and higher summer demand in Australia, the January–February trough likely reflects post-holiday spending normalisation combined with back-to-school budget pressure.

These seasonal patterns may help inform promotional planning and inventory management for future category strategies.

In [166...

```

merged_data.to_csv('merged_data.csv', index=False)
segment_metrics.to_csv('segment_metrics.csv')

```

Key Findings and Recommendations

Key Findings

- **Older Families (Budget)** generated the highest total chip sales and showed the highest purchase frequency (~5 transactions per customer), with the largest basket sizes — driven by strong multi-pack purchasing behaviour.

- **Mainstream Young Singles/Couples** represent the largest customer group by volume and pay the highest average price per unit, indicating a preference for premium products over value deals.
- **Multi-pack purchasing** is common across all segments but peaks in family segments, particularly Older Families, supporting their outsized contribution to total sales.
- **Kettle dominates** overall brand sales and is especially strong among Mainstream Young Singles/Couples. Budget Older Families distribute spend more evenly across Smiths and Doritos.
- **Seasonal trends** show a clear December sales peak driven by holiday and summer demand in Australia, with a notable February dip representing post-holiday normalisation.
- **Pack size preference** is consistent across all segments (~180g average), indicating pack size is not a meaningful differentiator between customer groups.

Recommendations

- Target family segments through value-focused promotions, larger pack sizes and multi-buy offers, as these customers contribute strongly through purchase frequency and basket size.
- Continue engaging Mainstream Young Singles/Couples using premium products, convenience-based offerings and new flavour innovations, as this segment shows higher willingness to spend per unit.
- Maintain strong visibility and promotional support for leading brands such as Kettle, Doritos and Smiths, as they are major contributors to overall category sales.

Conclusion

The analysis identified clear differences in purchasing behaviour across customer segments. Family households contribute strongly through frequent and higher-volume purchases, while younger mainstream customers contribute through higher-value purchasing behaviour. These insights can support more targeted promotional and category strategies for the upcoming chip category review.